

PeopleSoft Chatbot Install Guide (for ODA 19.10 or higher)

How to use this Document:

This document guides the application developer to create a basic skill, which displays user details and to integrate skill with PeopleSoft Using Chatbot Integration Framework. It advances further with the details of extending it to build a fully functional Bot as per the user requirement. The document revolves mainly around two areas

- 1) Oracle Digital Assistant (ODA) setup to enable PSFT ChatBot skills.
- 2) PeopleSoft Setup for Chatbot.

Note: If you are using ODA version on or before 19.04, please refer to PeopleSoft-ChatBot install guide for ODA 19.04 or lower.

Before you Get Started:

- Procure the required Oracle Cloud Account and configure the Oracle Digital Assistant. (for more details refer: <https://docs.oracle.com/en/cloud/paas/digital-assistant/service-administration/get-started-oda.html#GUID-1F76B688-9A9A-4870-BB7D-764ED0C594DD>)
- You have License for the Required PeopleSoft products and Integration Broker is setup and running
- Minimum PeopleTools and Apps release as below:

Pillar	PUM Release	Bugs (PT 8.57.07 and above)	Bugs (PT 8.58.02 and above)
HCM	34	30904061	30904061, 30885059
FSCM	35	30900860	30900860, 30885039
CRM	19	30975470	30975470, 30975771
ELM	21	30949218	30949218, 30954878
CS	17	30974571	30974571, 30975802

* For Chatbot Integration to work with People Tools Release 8.58.01, it is required to apply PeopleTools POC on the target environment.

Assumptions:

- 1) PeopleSoft Application Service is accessible on the open internet for the ODA Cloud instance to consume.
- 2) PeopleSoft holds a certificate signed by a valid CA and not a self-signed certificate.
- 3) Integration Broker is configured, up and running.
- 4) For the ease of documentation, we have taken “HCM” as the PeopleSoft application. However it can be extended for other pillars as well.
- 5) A user to authenticate PeopleSoft web services “**PSFTPROXY**” is created. Administrator may use appropriate user id and password based on your preference.
- 6) WebserverName : “webserverName.com”
- 7) Node: “nodeName”
- 8) Port Number: 8001

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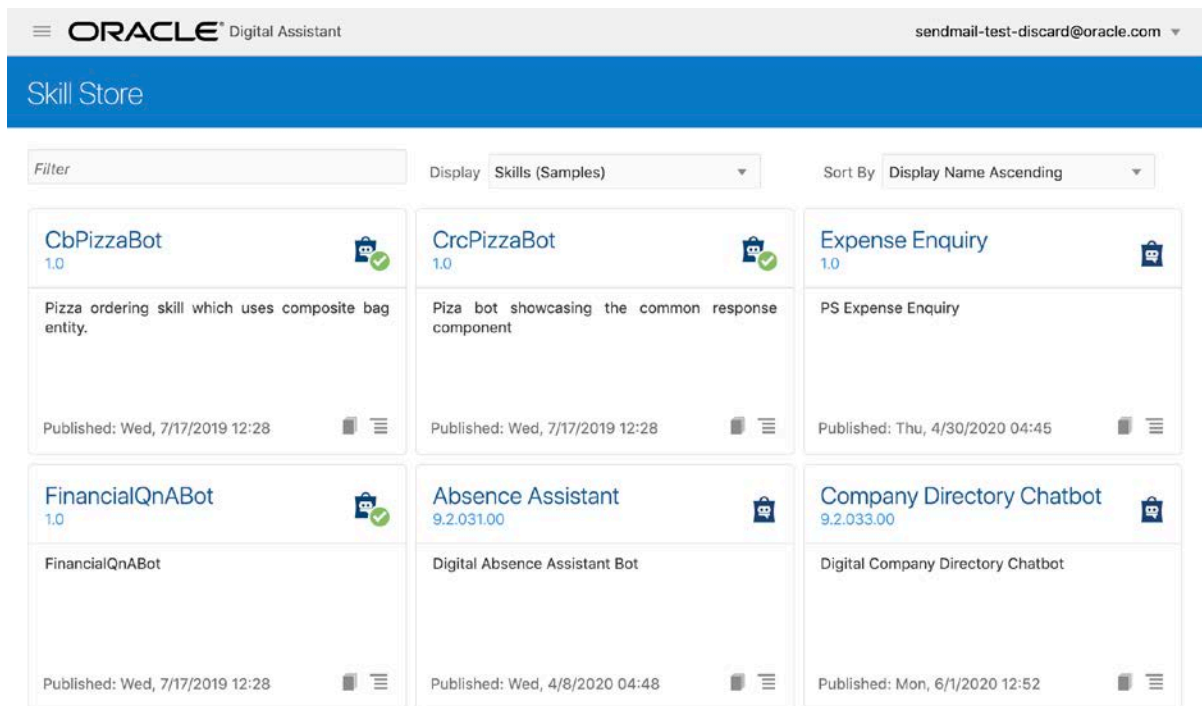
1 ODA Setup

1.1 Install and Configure the Skill

1.1.1 Sign In to Your ODA Instance from the Oracle Cloud Website.

1.1.2 Navigate to the Skill Store, click  to open the side menu. Select Development > Store.

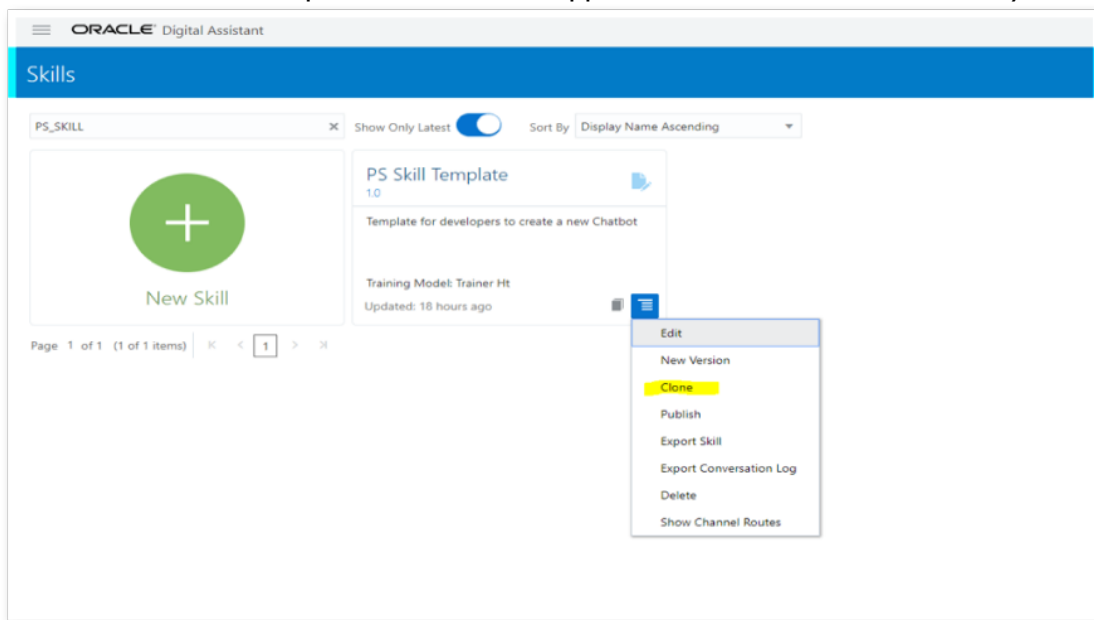
1.1.3 Install or Pull the latest PS Skill Template from Skill Store.



The screenshot shows the Oracle Digital Assistant Skill Store interface. At the top, there's a header with the Oracle logo and "Digital Assistant" text, and a user email "sendmail-test-discard@oracle.com". Below the header is a blue bar labeled "Skill Store". The main area contains a grid of skill cards. Each card displays the skill name, version, a brief description, and the publish date. The skills listed are:

Skill Name	Version	Description	Published Date
CbPizzaBot	1.0	Pizza ordering skill which uses composite bag entity.	Wed, 7/17/2019 12:28
CrcPizzaBot	1.0	Piza bot showcasing the common response component	Wed, 7/17/2019 12:28
Expense Enquiry	1.0	PS Expense Enquiry	Thu, 4/30/2020 04:45
FinancialQnABot	1.0	FinancialQnABot	Wed, 7/17/2019 12:28
Absence Assistant	9.2.031.00	Digital Absence Assistant Bot	Wed, 4/8/2020 04:48
Company Directory Chatbot	9.2.033.00	Digital Company Directory Chatbot	Mon, 6/1/2020 12:52

- 1.1.4 Clone the installed PS Skill Template and save it with a name of your choice (Cloning action is recommended for Template Skill while the Application Skills should be extended).



Create Clone

Display Name *
MyFirstSkill

Name *
MyFirstSkill

Version *
1.0

One-Sentence Description
My First Skill to test Peoplesoft Chatbot

☒ Open cloned skill bot afterwards

Clone

- 1.1.5 Open the Dialog Flow of the cloned skill. Replace the entire content with the below YAML:

```
#metadata: information about the flow
# platformVersion: the version of the bots platform that this flow was written to work with
metadata:
  platformVersion: "1.0"
main: true
name: PS_Skill_Template
#context: Define the variables which will used throughout the dialog flow here.
context:
  variables:
```

```

  UserID: "string"
  UserEmail: "string"
  UserEmplID: "string"
  Userlanguage: "string"
states:
start:
  component: "HCM.ps.GetSystemVariables"
properties:
  OperatorId: "UserID"
  EmailAddress: "UserEmail"
  EmployeeId: "UserEmplID"
  Language: "Userlanguage"
transitions:
  actions:
    SUCCESS: "displayUserDetails"
    perror: "connectionError"
displayUserDetails:
  component: "System.Output"
properties:
  text: "The Details of the current user are \n UserID: ${UserID} \n Email: ${UserEmail} \n EmployeeId: ${UserEmplID} \n Language:"
transitions:
  return: "done"
connectionError:
  component: "System.Output"
properties:
  text: "We are unable to establish connection with PeopleSoft. Please validate the configurations"
transitions:
  return: "done"

```

< Skills • MyFirstSkill DRAFT • 1.0

Validate
 Train

+ Components
?

Theme Default

```

1 #metadata: information about the flow
2 # platformVersion: the version of the bots platform that this flow was written to work with
3 metadata:
4   platformVersion: "1.0"
5   main: true
6   name: PS_Skill_Template
7   #context: Define the variables which will used throughout the dialog flow here.
8   context:
9     variables:
10      UserID: "string"
11      UserEmail: "string"
12      UserEmplID: "string"
13      Userlanguage: "string"
14   states:
15     start:
16       component: "HCM.ps.GetSystemVariables"
17       properties:
18         OperatorId: "UserID"
19         EmailAddress: "UserEmail"
20         EmployeeId: "UserEmplID"
21         Language: "Userlanguage"
22       transitions:
23         actions:
24           SUCCESS: "displayUserDetails"
25           perror: "connectionError"
26       displayUserDetails:
27         component: "System.Output"
28         properties:
29           text: "The Details of the current user are \n UserID: ${UserID} \n Email: ${UserEmail} \n EmployeeId: ${UserE
30           mplID} \n Language:"
31         transitions:
32           return: "done"
33       connectionError:
34         component: "System.Output"
35         properties:
36           text: "We are unable to establish connection with PeopleSoft. Please validate the configurations"
37         transitions:
38           return: "done"

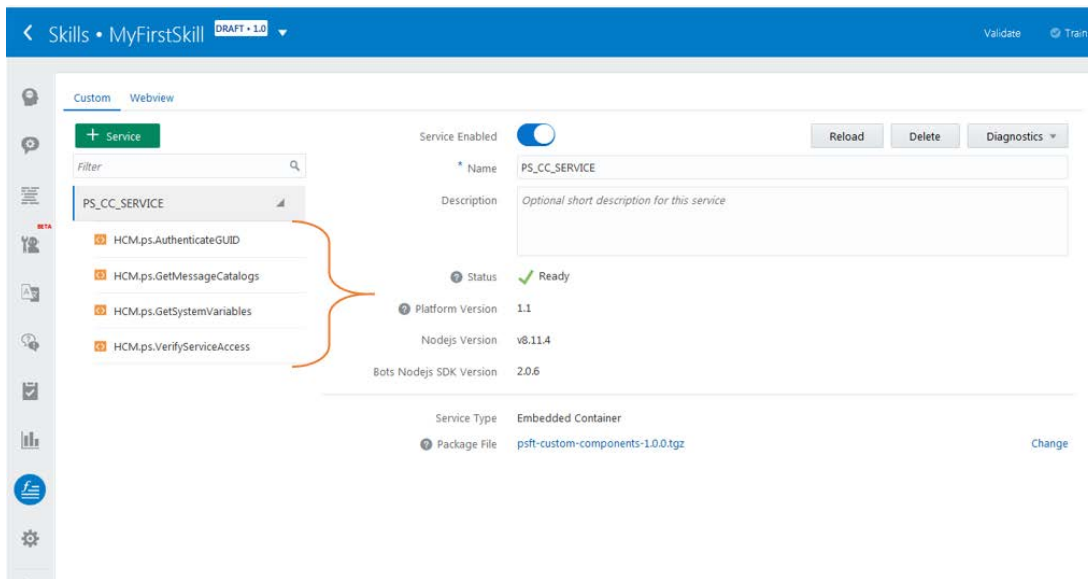
```

Variable Declaration

Invoking Delivered Custom Component

Displaying Results to the User

1.1.6 Navigate to Services tab , reload the service and ensure that the delivered custom components are listed.



Note: The name of the custom component will be prefixed with Application name. For e.g. The component name will be “CRM.ps.AuthenticateGUID” for CRM

1.1.7 Navigate to Skill Setting  > Configuration tab. Set the following values for:

1. PSHCMbaseurl: It denotes the base url of application service. Replace with actual server, port and nodename.

https://<WebserverName.com>:<port>/PSIGW/RESTListeningConnector/<nodeName>/PTCB_APPL_SVC.v1

2. PSHCMuserid: Set the peoplesoft proxy user who has access to invoke application services. To start with, Let us leave it, as it is (PSFTPROXY).

3. PSHCMpassword: It denotes the password for Proxy user. To start with, Let us leave it, as it is (PSFTPROXY).

Confidence Threshold: 0.6
The minimum confidence score required to match a skill's intent with user input. If there is no match, the transition action is set to 'unresolvedIntent'. (Minimum value 0, maximum value 1)

* Confidence Win Margin: 0.1
Only the top intent that exceeds the confidence threshold is picked if its confidence score exceeds that of other intents by this value or more. If other intents that exceed the confidence threshold have scores that are within that of the top intent by less than the win margin, these intents are also presented to the user. (Minimum value 0, maximum value 1)

* Unexpected Error Prompt: Oops I'm encountering a spot of trouble. Please try again later...
The message when there is an unexpected error

* Max States Exceeded Error Prompt: Your session appears to be in an infinite loop.
The message when the Bot appears to be in an infinite loop

* Expired Session Error Prompt: Your current chat session has expired. Please start a new chat with a new enquiry or request
The message when the session has expired

* OAuth Cancel Prompt: Authentication canceled.
The message when OAuth authorization is canceled

* OAuth Success Prompt: Authentication successful: You can return to the conversation.
The message when OAuth authorization succeeds

Custom Parameters

+ New Parameter

Filter parameters

Name	Display Name	Type	Value	Description
PSHCMbaseurl	HCM Base URL	String	https://<server>:<port>/PSIGW/REST/ListeningConnector/<node>/PTCB_APPI_SVC.v1	
PSHCMpassword	HCM Password	String	PSFTPROXY	
PSHCMuserid	HCM User ID	String	PSFTPROXY	

Page: 1 of 1 (1-3 of 3 items)

1.1.8 Validate it to ensure that the Skill is built without any errors.

< Skills • MyFirstSkill DRAFT • 1.0

Validate Train

✓ We didn't find any problems with the MyFirstSkill skill bot.

```

1 #metadata: information about the flow
2 # platformVersion: the version of the bots platform that this flow was written to work with
3 metadata:
4   platformVersion: "1.0"
5 main: true
6 name: PS_SKILL_TMPL
7 #context: Define the variables which will used throughout the dialog flow here.
8 context:
9   variables:
10     UserID: "string"
11     UserEmail: "string"
12     UserEmplID: "string"
13     Userlanguage: "string"
14
15 states:
16   start:
17     component: "HCM.ps.GetSystemVariables"
18     properties:
19       OperatorId: "UserID"
20       EmailAddress: "UserEmail"
21       EmployeeId: "UserEmplID"
22       Language: "Userlanguage"
23     transitions:
24       next: "displayUserDetails"
25   displayUserDetails:
26     component: "System.Output"
27     properties:

```

1.1.9 Read the below ODA documentation to know about intents, utterances, entites, dialog flow.

<https://docs.oracle.com/en/cloud/paas/digital-assistant/use-chatbot/using-oracle-digital-assistant.pdf>. Although our skill –“MyFirstSkill” doesn’t require those to start with, however It is mandatory to have an understanding on those concepts for building a fully functional bot.

1.1.10 For more details on Chat that bot integration framework refer [PeopleSoft Chatbot Red Paper](#)

1.2 Create Channel

1.2.1 Navigate to Development>Channels.

1.2.2 Create new Channel with the following details:

Name: ChannelForMySkill

Channel Type: Oracle Web

Allowed Domains: * (Specify the actual domain name in the production environment)

Client Authenticaiton Enabled: Disable

Session Expiration(Minutes): Default Value

The screenshot shows a 'Create Channel' modal window. The fields are as follows:

- Name:** ChannelForMySkill
- Description:** Optional short description for this channel
- Channel Type:** Oracle Web
- Allowed Domains:** *
- Client Authentication Enabled:** Disabled (toggle switch)
- Session Expiration (minutes):** 60 (with up/down arrows and a 'Default' label)

A green 'Create' button is located at the bottom right of the modal.

Users

Agent Integrations

DA as Agent

Applications

System

Channel

Filter

Absence_Assistant_PM

Benefit_H92LLQA1

channel9

Channel_for_CRM_Chatbot_1

ChannelForMySkill

Chatbot_SKM

CHATBOTDEMO_SAVITHA

CIF_DEV

CIF_DEV_Attachment

company_directory_channel

Route To

MyFirstSkill

DRAFT • 1.0

Channel Enabled

☒

Reset Sessions

Name

ChannelForMySkill

Description

Optional short description for this channel

Channel Type

Oracle Web

Allowed Domains

*

Secret Key

ULQ7xH3c0u6s8m5x0SNa1s1a2F

Reset

Channel Id

9c03e4dfe2d83d0f5d388e0772d68067

Client Authentication

☐

Enabled

Session Expiration

60

Default

2 PIA Setup

2.1 Proxy User Creation

2.1.1 Log in to PIA as Admin User.

2.1.2 Create a User “PSFTPROXY” with user ID and required password and assign the roles EOCB Service user and PTCB_USER.

The screenshot shows the 'User Profiles' page in the 'Employee Self Service' application. The 'General' tab is selected. The user ID is 'PSFTPROXY'. The 'Logon Information' section includes a 'Symbolic ID' dropdown set to 'SYSADM1', checkboxes for 'Change Password?' and 'Password Expired?', and a 'User ID Alias' field set to 'PSFTPROXY'. The 'General Attributes' section includes a 'Language' dropdown set to 'English', a 'Currency' dropdown, a 'Default Mobile Page' search field, and a checkbox for 'Enable Expert Entry'. The 'Permission Lists' section includes search fields for 'Navigator Homepage', 'Process Profile', 'Primary', and 'Row Security'. There are also links for 'Edit Email Addresses' and 'Instant Messaging Information'.

The screenshot shows the 'User Profiles' page in the 'Employee Self Service' application, with the 'ID' tab selected. The user ID is 'PSFTPROXY'. The 'ID Types and Values' section includes a search bar, a dropdown for '*ID Type' set to 'None', and a table with columns 'Attribute Name', 'Attribute Value', and 'Description'. The 'User Description' section includes a 'Description' search field and a link 'Set Description or type in User Description'. At the bottom, there are buttons for 'Save', 'Return to Search', 'Add', and 'Update/Display'. A breadcrumb trail at the bottom reads: 'General | ID | Roles | Workflow | Audit | Links | User ID Queries'.

Employee Self Service

User Profiles

General

ID

Roles

Workflow

Audit

Links

User ID Queries

User ID

PSFTPROXY

Description

Dynamic Role Rule

Execute on Server

Process Monitor

Service Monitor

Test Rule(s)

Refresh

Execute Rule(s)

User Roles

1-2 of 2

View All

Role Name	Description	Dynamic		View Definition		
EOCB Service User	Anonymous Service User	☐	Route Control	View Definition	+	-
PTCB_USER	ChatBot User	☐	Route Control	View Definition	+	-

Save

Return to Search

Add

Update/Display

2.2 Configure Bot ID

2.2.1 Navigate to Enterprise Components > Chatbot Configurations > Bot Definition page.

2.2.2 Set the ODA Release Version as – On or After 19.10 and enter the ODA uri and not the complete ODA url

For the ODA URL: <https://oda-XXXXXXXXXXXXassistant.oci.oc-test.com/botsui>

ODA Server URI is oda-XXXXXXXXXXXXassistant.oci.oc-test.com

Manager Self Service

Bot Definitions

39

Save

General Configurations

ODA Release Version

On or After 19.10

*Default Branding Template

Default Branding – ODA

*ODA Server URI

oda-XXXXXXXXXXXXassistant.oci.oc-test.com

* Changing the ODA Release Version value will remove the branding for earlier version, associated with one or more bot definitions. The default branding template for the new ODA release version value will be applied for the bot definitions and can be manually overridden on bot definition page.

Bot Definitions

+ Add

2 rows

Bot ID	Bot Name	Bot Type	Edit	Delete
1	HGA_CHAT_ASST	Absence Assistant	Both	
2	HRCD_CHAT_ASST	Employee Directory Assistant	Both	

2.2.3 Click Add to open the Add Bot Definition Page.

Employee Self Service

Add Bot Definition

Bot ID: NEW_SKILL

Add Cancel

Bot Identifier: NEW_SKILL

*Bot Name: <Skill Name>

*Bot Channel ID: <channel id copied from ODA channel>

*Bot Type: Both

*Branding Template:

Business Branding

Business Name:

Business Icon:

Roles

*Role Name
1 <Role>

Set the value for following fields on the above Bot Definition page:

- Bot Name: Name of the bot and it can be any name as it is contextually valid only within PeopleSoft.
- Bot Channel ID: Channel ID which was copied from the [channel](#)
- Bot Type: Set its value as Both or Web based.
- Role Name: The role which the user must have to access the Bot. Let us set it is “EOCB Client User” for the skill “MyFirstSkill”. Ensure that the chatbot user has this role.

Add Bot Definition

Maintain Bot Definition

Save

Bot Identifier: NEW_SKILL

*Bot Name: My First Skill

*Bot Channel ID: 5799%-xxxxxxxxxxxxxxxxx8989f

*Bot Type: Both

*Branding Template:

Business Branding

Business Name:

Business Icon:

Roles

*Role Name
1 EOCB Client User

2.3 Configuration for Launching Chatbot from Tile

2.3.1 Navigation People Tools> Portal> Structure and Content.

2.3.2 Click on Add Content Reference.

Link	Node Name	Label	Edit	Sequence number	Create Link	Number of links
HRMS	Company Directory		Edit	0	Create Link	0
HRMS	New_Skill		Edit	0	Create Link	0
HRMS	Performance Process		Edit	20	Create Link	0
HRMS	Careers		Edit	100	Create Link	0
HRMS	Careers		Edit	110	Create Link	0
HRMS	Company Directory		Edit	200	Create Link	0
HRMS	Company Directory (Redirect)		Edit	205	Create Link	0
HRMS	Company Directory Tree Detail		Edit	210	Create Link	0
HRMS	Search-Company Directory		Edit	300	Create Link	0
LOCAL_NODE	Classic Home		Edit	9978	Create Link	0
LOCAL_NODE	Fluid Home		Edit	9978	Create Link	0
LOCAL_NODE	Usage Monitoring		Edit	9979	Create Link	0
LOCAL_NODE	Change My Password		Edit	9980	Create Link	0
LOCAL_NODE	My Personalizations		Edit	9983	Create Link	0
LOCAL_NODE	My System Profile		Edit	9985	Create Link	0
LOCAL_NODE	My Dictionary		Edit	9989	Create Link	0
LOCAL_NODE	My Feeds		Edit	9992	Create Link	0

[Add Content Reference](#) [Add Content Reference Link](#)

2.3.3 Create new content reference.

Set the url type as PeopleSoft Generic URL and specify the below url in Portal URL field. Note that Bot ID created in above step (2.2.3) is appended as query string parameter to the Portal URL.

c/EOCB_MENU_FL.EOCB_CLIENT_FL.GBL?BOTID=NEW_SKILL

[Root > Self Service > HR Fluid Components >](#)

Content Ref Administration

CreatedBy PS
Parent Folder HR Fluid Components
[Copy object](#)
[Select New Parent Folder](#)

Name EOCB_NEWSKILL_FL_GBL
*Label New_Skill

Long Description (254 Characters)

Product

Sequence number

Owner ID

Usage Type Target

Storage Type Remote by URL

*Valid from date 12/03/2019
Valid to date
Creation Date 12/03/2019
☐ WSRP Producioble
☒ No Template
☒ Fluid Mode
☒ Display on Small Form Factor

[Create Content Reference Link](#) [Add Content Reference](#) [Test Content Reference](#)

URL Information

Node Name HRMS

URL Type PeopleSoft Generic URL

*Portal URL c/EOCB_MENU_FL.EOCB_CLIENT_FL.GBL?BOTID=NEW_SKILL

GeneralSecurityFluid Attributes

Root > Self Service > HR Fluid Components >

Content Reference Security

Label: New_Skil

Public

Author Access

Security Authorizations

Q

1-1 of 1

Type	Name	Description	View Definition		
1Permission List	EOCB_CLIENT_USR	EOCB Client User	View Definition	+	-

Inherited Security Authorizations

Q

1-1 of 1

Type	Name	Description	View Definition
			View Definition

Save

Notify

General | Security | Fluid Attributes

GeneralSecurityFluid Attributes

Root > Self Service > HR Fluid Components >

Title Information

Image NamePS_CHAT_CLIENT_M_FL

Display InModal

Modal ParameterssStyle@frame-pt_chatclientpage.bAutoClose@0

Height1

Width1

Refresh Timer0

Event Name

Interactive

Disable Main Hotspot

Autosize

Resizable

Dynamic Tile Content

URL TypeNone

Title Repository

Select a folder below to create a link to the Tile Repository. If the folder selection is disabled, this Tile is already available in the Tile Repository.

FolderHC_SELF_SERVICE_FL_GBL

Folder LabelEmployee Self Service

[View Definition](#)

Include In Repository

☒ Homepage

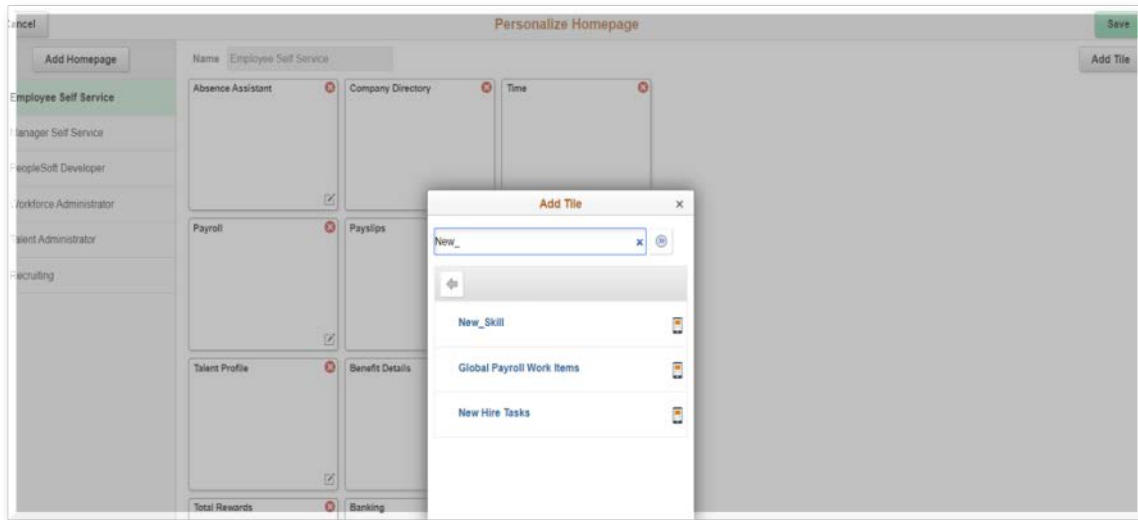
☒ Dashboard

☒ NavBar

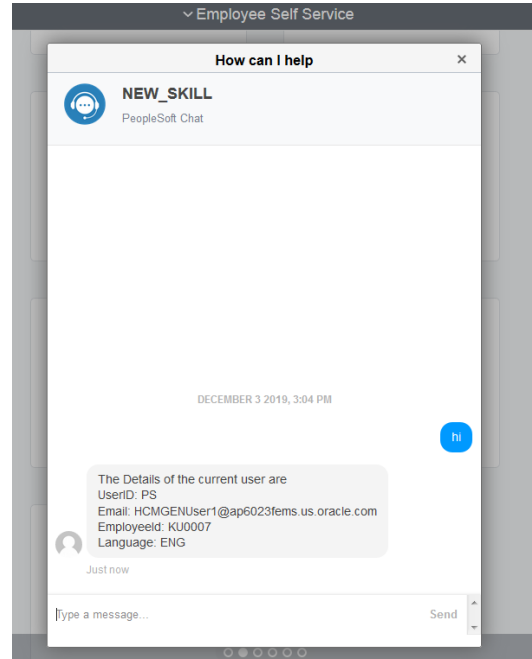
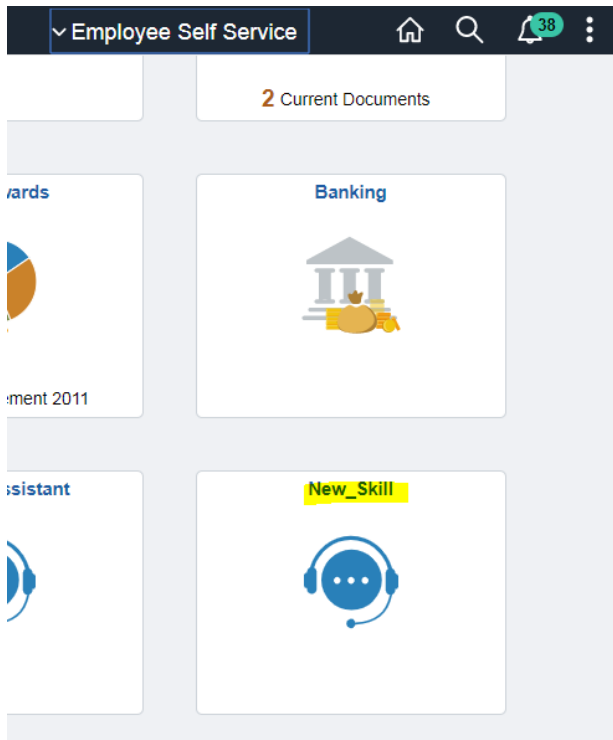
2.3.4 Add Tile to ESS Home Page.

Navigate to Homepage > Select Personalize Homepage.

Select Add Tile. Select Employee Self Service. Choose New_Skill and save it.

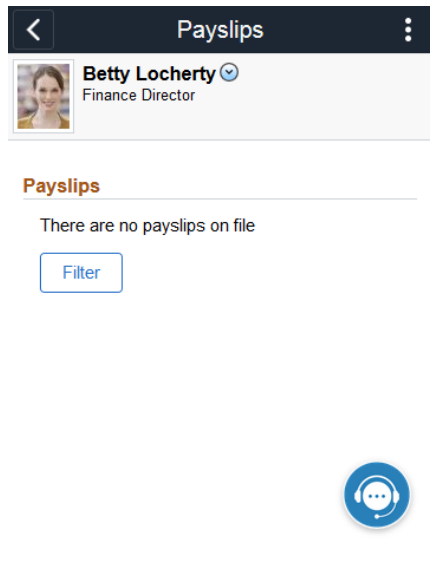


Navigate to Employee Self Service to Launch New_Skill



2.3.5 Setup to Access ChatBot from Embedded Icon

Now we will embed the Chatbot client widget on a component as shown below:



2.3.6 Navigate to Enterprise Components > Chatbot Configurations > Component Mapping.

2.3.7 Choose any fluid component. In this example we have taken the one for Payslip. Add a new component name and associate the Bot id to it.

The first screenshot shows the 'Add Component Mapping' dialog. It has a title bar 'Add Component Mapping'. Inside, there are two input fields: 'Component' with the value 'GP_SS_EE_PSLP_FLU' and a search icon, and 'Market' with a dropdown menu showing 'GBL'. At the bottom, there are two buttons: 'Add' (green) and 'Cancel' (grey).

The second screenshot shows the 'Maintain Component Mapping' interface. It has a title bar 'Maintain Component Mapping' and a back arrow labeled 'Component Mapping'. The main content area shows 'Component GP_SS_EE_PSLP_FLU' and 'Market GBL'. Below this, there is a section titled 'Associated Bots'. It contains a table with the following data:

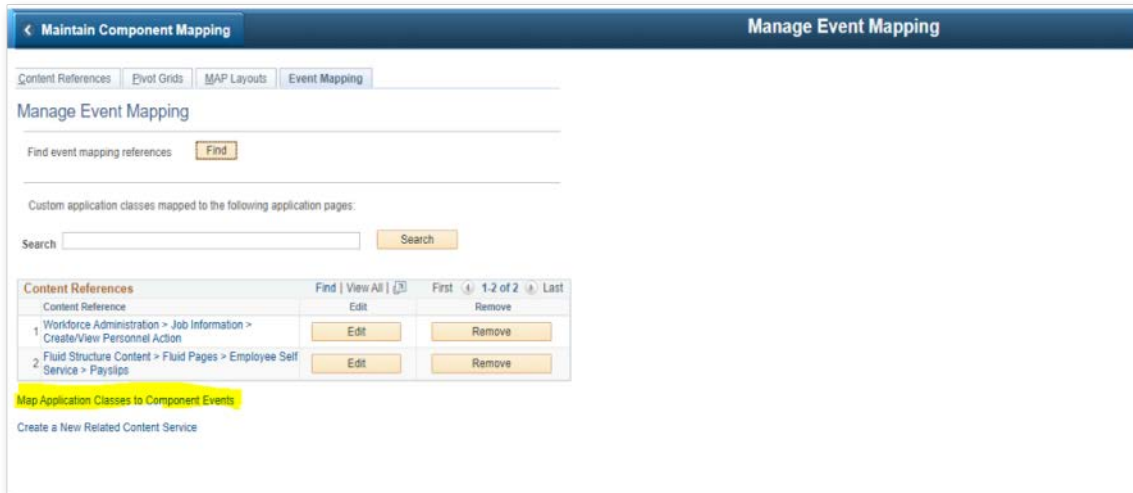
Order	Bot ID
1	NEW_SKILL

At the bottom right of the table, there are two buttons: '+' and '-'.

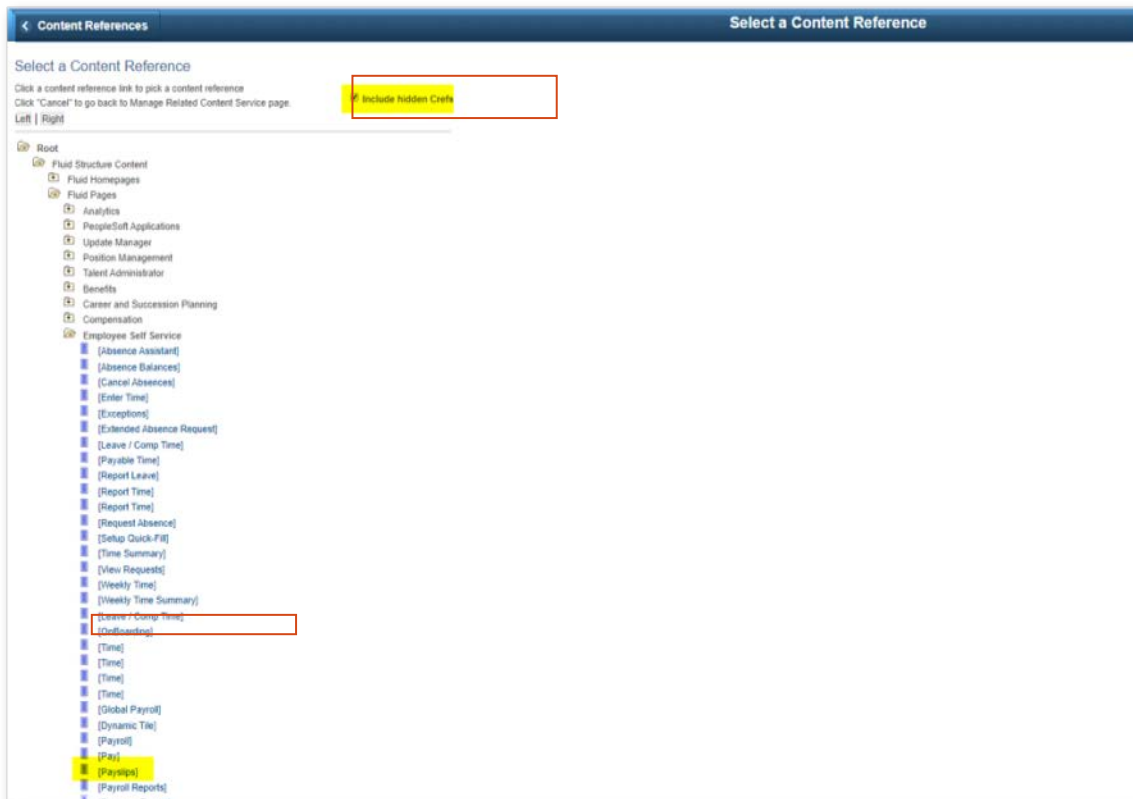
2.3.8 Navigate to People Tools > Portal> Related Content Service > Manage Content Reference.

2.3.9 Navigate to Event Mapping tab.

2.3.10 Select Map Application Classes to Component Events.



2.3.11 Select the path Fluid structure content > Fluid pages > ESS> Payslips (this is for configuring the chat client in the payslip page). Relate to 2.4.2 that we should use the navigation of the component used there.



Content References

Configure Event Mapping

Event Mapping

Configure Event Mapping

Configure the custom application class with the selected event. Custom PeopleCode can be configured to be executed before (pre process) or after (post process) defined for the same event from the component definition.

Content Reference [Payslips](#)

Component Level Event Mapping

Map the custom application class to the Component level event.

General

Personalize | Find | View All | 1 of 1 | First | Last

Enable	Event Name	Select	Service ID	Sequence Number	Processing Sequence
☒	PostBuild		EOCB_CHATCLIENT		Post Process

Page Level Event Mapping

Map the custom application class to the Page level event

General

Instance Information

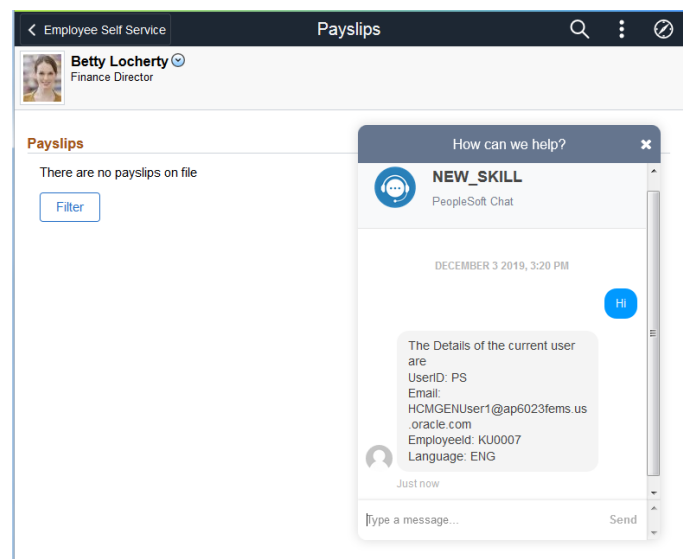
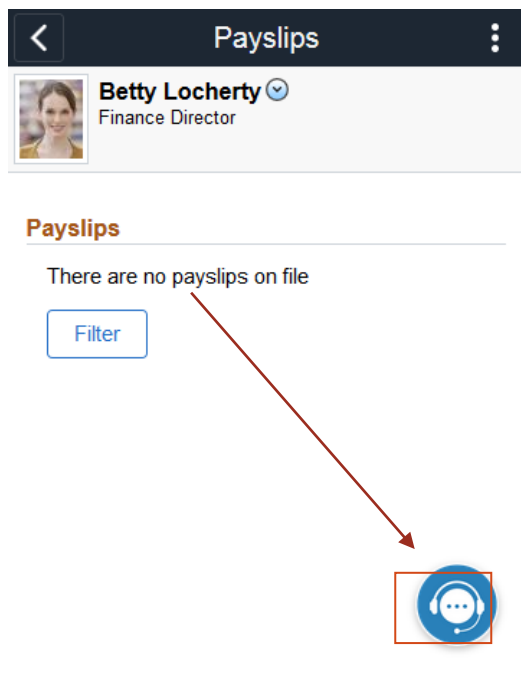
Personalize | Find | View All | 1 of 1 | First | Last

Enable	Unrestricted Prompt	Page	Event Name	Select	Service ID	Sequence Number	Processing Sequence
☒	☐		Activate				

Component Record Level Event Mapping

Map the custom application class to the Component Record level event

2.3.12 Navigate to Employee Self Service > Payslips (where embedded chat windows is configured). Click on the embedded Chabot icon and launch the chat window.



3 Adding Custom Components.

The Skill PS_SKILL_TMPL comes with delivered custom components. In order to build a functionally comprehensive Skill, there would arise a need to add more custom components. This section will take you through the steps for adding custom components within the Chabot Integration Framework.

Let us try to create a new custom component which accepts the userID as the input parameter and returns the details about the user such as name, emailID and Employee ID.

3.1 Creating Services in Application Service Framework.

3.1.1 Build Application Classes.

- Create an Application package “HCM_CHATBOTSERVICES”
- Create and insert the Application class “GetEmployeeDetails” to the App package.
- Ensure that this application class extends the class PTCBAPPLSVCDEFN:ApplicationServiceBase and implements the method invokeService.

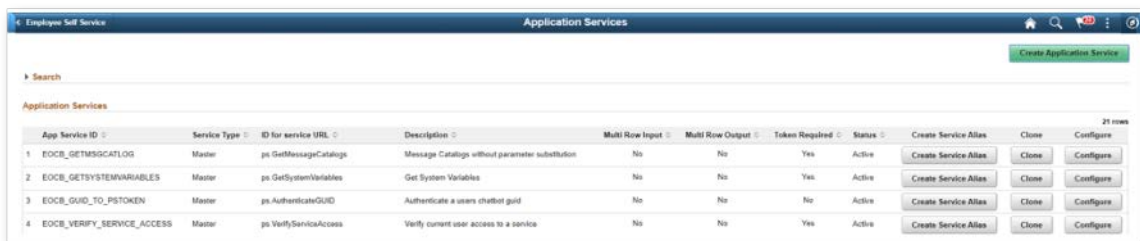
CHATBOTSERVICES : GetEmployeeDetails
<pre>import PTCBAPPLSVCDEFN:ApplicationServiceBase; class GetEmployeeDetails extends PTCBAPPLSVCDEFN:ApplicationServiceBase method invokeService(); property boolean IsError; property string ErrorMsg; end-class; method invokeService /* Extends/implements PTCBAPPLSVCDEFN:ApplicationServiceBase.invokeService */ Local string &name, &emailid, &emplid, &userID; /****Fetching value of input parameters*/ &userID = %This.ServiceAPI.getInputParameter("userID"); SQLExec("SELECT oprdefndesc,emplid,emailid from psoprdefn where OPRID=:1", &userID, &name, &emailid, &emplid); If (All(&name)) Then /****Setting Values of Ooutput parameters*/ %This.ServiceAPI.setOutputParameter("name", &name); %This.ServiceAPI.setOutputParameter("emailID", &emailid); %This.ServiceAPI.setOutputParameter("emplID", &emplid); /****Setting Values of input arguments*/ %This.ServiceAPI.ResultState = "Success"; Else %This.ServiceAPI.ResultState = "Failed"; End-If; end-method;</pre>

3.1.2 Login to PIA as Admin user.

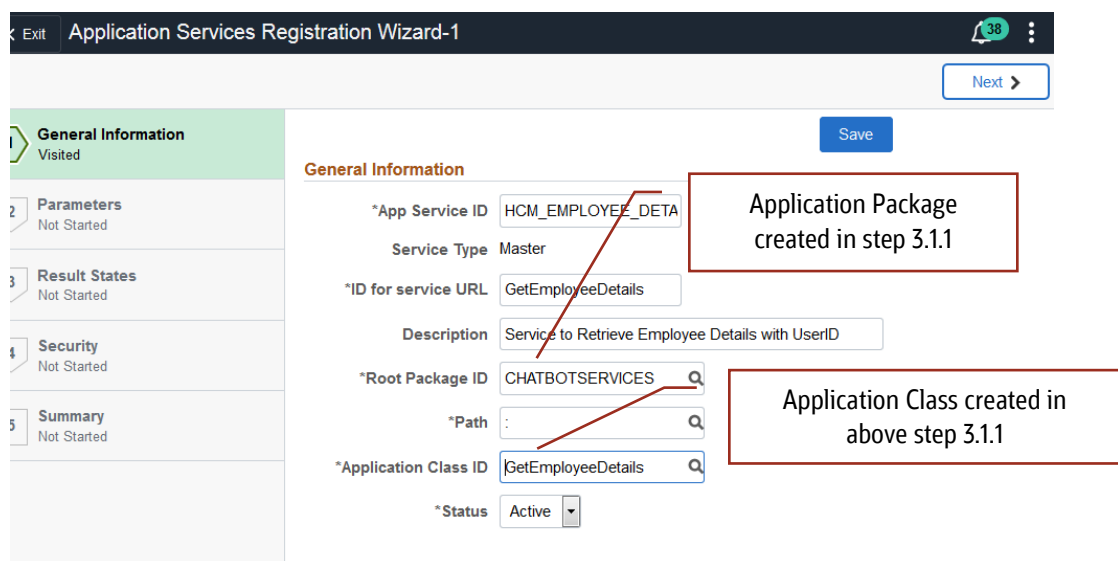
3.1.3 The Create Application Service button will launch the registration wizard for registering a new Application Service.

Navigation: Main Menu > PeopleTools > Integration Broker > Web Services > Application Services > Application Services

Note: The user must have the role PTCB_ADMINISTRATOR to access the Application.



App Service ID	Service Type	ID for service URL	Description	Multi Row Input	Multi Row Output	Token Required	Status	Create Service Alias	Close	Configure
1 EOCB_GETMSGCATLOG	Master	ps.GetMessageCatalogs	Message Catalogs without parameter substitution	No	No	Yes	Active	Create Service Alias	Close	Configure
2 EOCB_GETSYSTEMVARIABLES	Master	ps.GetSystemVariables	Get System Variables	No	No	Yes	Active	Create Service Alias	Close	Configure
3 EOCB_GUID_TO_PSTOKEN	Master	ps.AuthenticateGUID	Authenticate a users chatbot guid	No	No	No	Active	Create Service Alias	Close	Configure
4 EOCB_VERIFY_SERVICE_ACCESS	Master	ps.VerifyServiceAccess	Verify current user access to a service	No	No	Yes	Active	Create Service Alias	Close	Configure



General Information

*App Service ID: HCM_EMPLOYEE_DETAILS
Service Type: Master
*ID for service URL: GetEmployeeDetails
Description: Service to Retrieve Employee Details with UserID
*Root Package ID: CHATBOTSERVICES
*Path: /
*Application Class ID: GetEmployeeDetails
*Status: Active

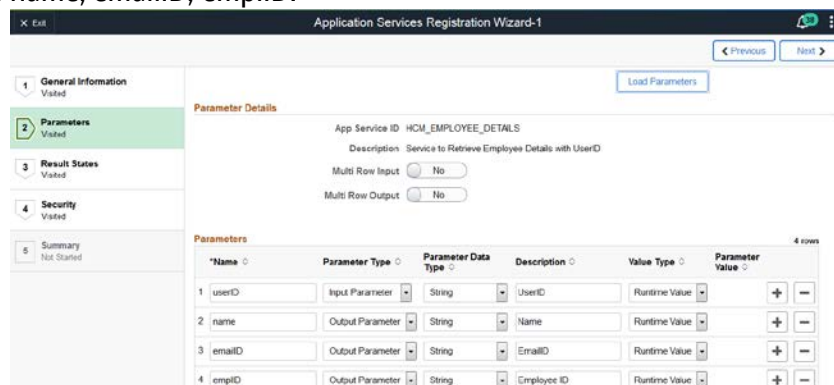
Application Package created in step 3.1.1

Application Class created in above step 3.1.1

Set the input and output parameter details.

Input Parameter: userID

Output Parameter: name, emailID, emplID.



Parameter Details

App Service ID: HCM_EMPLOYEE_DETAILS
Description: Service to Retrieve Employee Details with UserID
Multi Row Input: No
Multi Row Output: No

Name	Parameter Type	Parameter Data Type	Description	Value Type	Parameter Value
userID	Input Parameter	String	UserID	Runtime Value	
name	Output Parameter	String	Name	Runtime Value	
emailID	Output Parameter	String	EmailID	Runtime Value	
emplID	Output Parameter	String	Employee ID	Runtime Value	

Set the possible result states of invoking the service.

Application Services Registration Wizard-1

Exit Previous Next

1 General Information Visited

2 Parameters Visited

3 Result States Visited

4 Security Not Started

5 Summary Not Started

Result State Information

App Service ID HCM_EMPLOYEE_DETAILS

Description Service to Retrieve Employee Details

Load Result States

Result States 2 rows

*App Service States	Description	*Result Category		
1 Success	Success	Success	+	-
2 Failure	Failure	Failure	+	-

Save and submit to conclude the Service Registration Process.

Application Services Registration Wizard-1

Exit Previous Next

1 General Information Visited

2 Parameters Visited

3 Result States Visited

4 Security Visited

5 Summary Not Started

Security Details

App Service ID HCM_EMPLOYEE_DETAILS

Description Service to Retrieve Employee Details with UserID

Token Required Yes

Security 1 row

Permission List	Permission List Description		
1 EOCB_CLIENT_USER	EOCB Client User	+	-

3.2 Updating Custom Component Package.

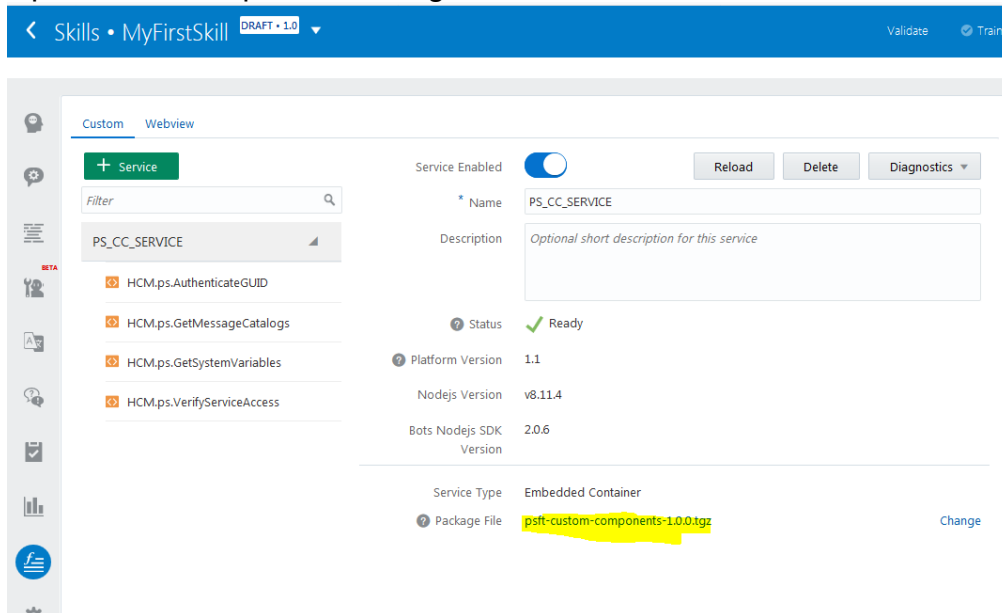
- 3.2.1 In order to verify the new application service is available in the application services, try accessing the Application Service URL appended by “/describe” and see if the details of the newly created service are listed out as per the configuration. When prompted for Login credentials, enter the credentials of the user with Permission List “PTCBAUTH”. In this case it is PSFTPROXY/PSFTPROXY. Copy the content and save it in as the json file - hcm_describe.json.

URL:

https://<WebserverName.com>:<port>/PSIGW/RESTListeningConnector/<nodeName>/PTCB_APPL_SVC.v1/describe

JSON	Raw Data	Headers
Save	Copy	
ParseStatus:	true	
Authorized:	true	
Services:		
0:	{...}	
1:	{...}	
2:	{...}	
3:	{...}	
4:		
ApplicationServiceID:	"HCM_EMPLOYEE_DETAILS"	
IDForServiceURL:	"GetEmployeeDetails"	
Description:	"Service to Retrieve Employee Details"	
CacheSupportLevel:	" "	
IsTokenMandatory:	false	
IsMultiRowInput:	false	
IsMultiRowOutput:	false	
InputParameters:		
0:		
ParameterName:	"userID"	
DataType:	"String"	
Description:	"UserID"	
OutputParameters:		
0:		
ParameterName:	"emailID"	
DataType:	"String"	
Description:	"EmailID"	
1:		
ParameterName:	"emplID"	
DataType:	"String"	
Description:	"Employee ID"	
2:		
ParameterName:	"name"	
DataType:	"String"	
Description:	"Name"	
ResultStates:		
0:		
Name:	"Failure"	
Category:	"Failure"	
Description:	"Failure"	
1:		

3.2.2 Open the Skill and click on the custom component package to download the custom component package `psft-custom-components-X.x.x.tgz`.



3.2.3 Extract the .tgz file to .tar and then extract the tar to the folder “package”.

package	12/3/2019 4:39 PM	File folder	
psft-custom-components-1.0.0.tar	8/26/2019 4:04 PM	TAR File	67 KB
psft-custom-components-1.0.0.tgz	8/26/2019 4:04 PM	TGZ File	11 KB

3.2.4 Open the folder `package\psft-lib\config` and replace the existing `hcm_describe.json` file with the new `hcm_describe.json` file (containing the new service) created in step 3.2.1. For FSCM it is `fscm_describe.json`.

package > psft-lib > config			
Share with ▾ New folder			
Name	Date modified	Type	Size
config.js	1/29/2019 1:16 AM	JScript Script File	1 KB
environments.json	5/30/2019 3:24 PM	JSON File	1 KB
fscm_describe.json	1/29/2019 1:16 AM	JSON File	3 KB
hcm_describe.json	5/31/2019 11:56 AM	JSON File	7 KB

3.2.5 Open `environments.json` and update environment name and `static_describe`.

```

{
  "HCM": {
    "baseUrl": "https://<server>:<port>/PSIGW/RESTListeningConnector/<node>/PTCB_A",
    "user": "PSFTPROXY",
    "password": "PSFTPROXY",
    "static_describe": "../config/hcm_describe.json",
    "rootca": ""
  }
}

```


3.2.6 Open the command prompt, change directory to the folder package and run following command:

```
npm install @oracle/bots-node-sdk
npm install

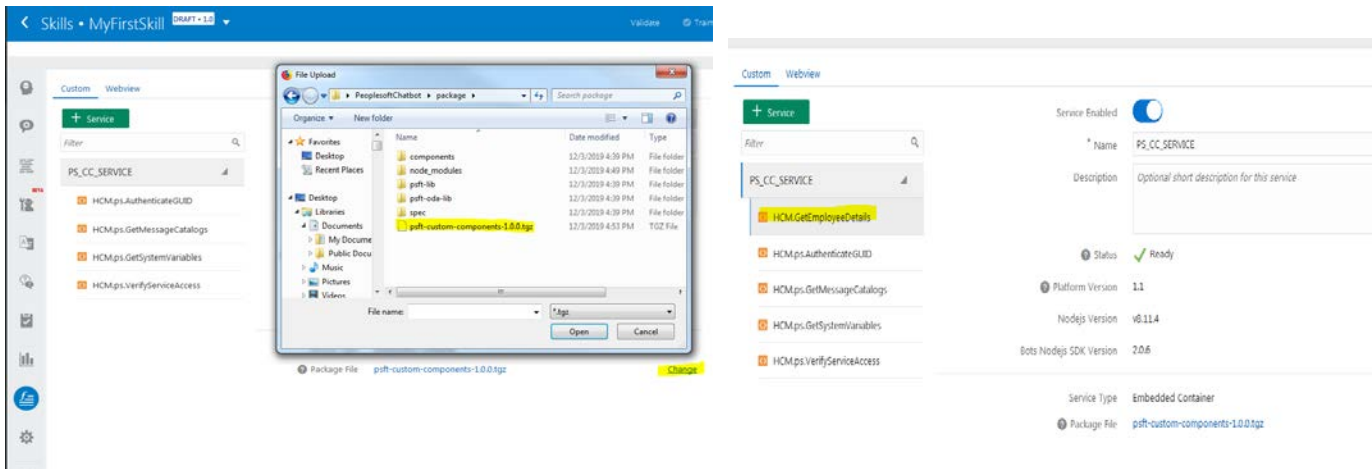
C:\Users\> cd package

C:\Users\> cd package
C:\Users\> npm install
npm notice created a lockfile as package-lock.json. You should commit this file.
npm WARN psft-custom-components@1.0.0 No repository field.
npm WARN psft-custom-components@1.0.0 No license field.
added 48 packages in 11.633s
```

3.2.7 Run the command npm pack to generate the updated custom component package file. The new file will be saved inside the package folder. The next time when you to generate a custom component package file with new changes, delete the existing file before running the npm pack command.

```
C:\Users\> cd package
C:\Users\> npm pack
> psft-custom-components@1.0.0 prepack C:\Users\> cd package
> npm run bots-node-sdk -- pack --dry-run .
> psft-custom-components@1.0.0 bots-node-sdk C:\Users\> cd package
> bots-node-sdk "pack" "--dry-run" "."
*****creating a component for service HCM.ps.GetMessageCatalogs
*****creating a component for service HCM.ps.GetSystemVariables
*****creating a component for service HCM.ps.AuthenticateGUID
*****creating a component for service HCM.ps.VerifyServiceAccess
-----
Component Package 'package' is valid!
-----
psft-custom-components-1.0.0.tgz
C:\Users\> cd package
```

3.2.8 Upload the new custom component package file in the skill services tab.



(In order to adhere to the naming convention it is recommend to prefix the Application Service name with ps while creating the service in Registration wizard)

3.2.9 Navigate to the dialog flow and replace it with following YAML.

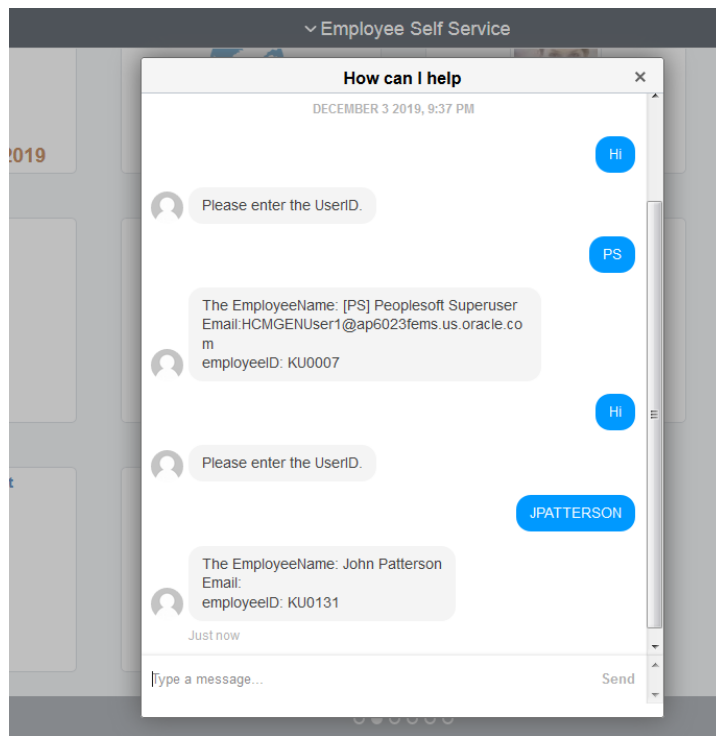
```
#metadata: information about the flow
# platformVersion: the version of the bots platform that this flow was written to work with
metadata:
  platformVersion: "1.0"
main: true
name: PS_SKILL_TMPL
#context: Define the variables which will used throughout the dialog flow here.
context:
  variables:

    employeeUserID: "string"
    employeeID: "string"
    employeeName: "string"
    employeeEmailID: "string"

states:
  getUserID:
    component: "System.Text"
    properties:
      prompt: "Please enter the UserID."
      variable: "employeeUserID"

  fetchEmployee:
    component: "HCM.GetEmployeeDetails"
    properties:
      userID: "${employeeUserID.value}"
      name: "employeeName"
      emailID: "employeeEmailID"
      emplID: "employeeID"
    transitions:
      actions:
        Success: "displayEmployeeDetails"
        pserror: "connectionError"
  displayEmployeeDetails:
    component: "System.Output"
    properties:
      text: " The EmployeeName: ${employeeName} \n Email:${employeeEmailID} \n employeeID: ${employeeID} "
    transitions:
      return: "done"
  connectionError:
    component: "System.Output"
    properties:
      text: "We are unable to establish connection with PeopleSoft. Please validate the configurations "
    transitions:
      return: "done"
```

3.2.10 Navigate to the PeopleSoft Employee Self Service homepage and launch the New_Skill to test the new custom component.



4 Troubleshooting Guide

4.1 Issues:

ID	Issues	Troubleshooting steps
I1	Chat client is spinning and not loading	Validate T1 a,b,c,d, T2 a, T8
I2	The chat client is not responding to the user's message.	Validate T2 a,b,c
I3	The chat client is responding with connection errors like "We are unable to establish connection with PeopleSoft. Please validate the configurations", "PS error occurred", ""	Validate T2a,7, T5,T6
I4	The chat client is responding with ODA configured error messages like "Oops encountered an unexpected error", "we are experiencing an unexpected error"	Validate T3,T4

4.2 Troubleshooting Steps:

ID	Description
T1	Bot Definition Page (Enterprise Components → Chatbot Configuration-> Bot definition) (a) Select the ODA version higher or equal to ODA 19.10. (b) Enter the ODA URI in the below format in Bot Definition Page and save it. oda-f8f3f3569b654XXXXXXXXXX-t0.data.digitalassistant.oci.oc-test.com (c) Edit the respective Bot and update the channel id. (d) The Bot has to be updated with appropriate roles and ensure that Chatbot user possess those roles to access the Bot.
T2	ODA Channel Configuration (a) Ensure that the channel is routed to the appropriate skill. (b) Turn off the client authentication. (c) Enter * in the allowed domain fields to allow request from all domains, or enter the peoplesoft domain to restrict it to PeopleSoft.
T3	Validate the ODA dialog flow and correct if there are any syntax errors in the YAML.
T4	Reload the custom component services and check if the all the components are loaded properly.
T5	Ensure the Proxy user (PSFTPROXY) has appropriate roles , reset the password and check if the alias name is appropriate.
T6	Set Pstoken/OAUTH Authentication. 1. Navigate to PeopleTools → Integration broker → Webservices → Application Services 2. Open PT_Configuration set the Authentication type as "PS ODA Authentication", if you are not using any identity management such as IDCS
T7	ODA Tester

1. Open Skill from on the ODA window and click on the Tester button on the Left pane , and test the skill

The screenshot displays the ODA Skill Tester interface. On the left, a chat window shows a user input 'Hi' and a system response 'PS Error Occured, Please contact the helpdesk'. The right pane shows the 'States' section with a flowchart for the 'DevTwilioPsToken Skill'. The flowchart consists of four steps: 'AuthLogin', 'DefaultGreetingFlag', 'GetUserProfile', and 'PSError'. Below the flowchart, the 'Variables' section lists several variables, including 'pserrorcode' which is highlighted in red and set to 'SystemError'.

pserrorcode	Possible Cause	Solution
SelfSignedCert UnverifiedLeafSignature or Other certificate related error	The PSHCMbaseurl in the skill parameter have https url but the PeopleSoft server carries a self-signed or an invalid certificate.	PeopleSoft server must have a certificate issued by a valid certifying authority.
GUIDAuthError	1. Try Launching the chat client from PeopleSoft and check if it works fine. If you are still getting the error refer the below steps: 2. You are using http base URL and the service operation PTCB_APPL_SVC is configured as "SSL". 3. The user doesn't have access to the application services. 4. Incorrect (or lower case) Node name or it is in lower case. 5. Incorrect (or lower case) application service name in the Base url configured in the Skill.	
401 Unauthorized	1. Incorrect Proxy user/password 2. Proxy user does not have necessary roles.	

	ENOTFound ETimedOUT	Incorrect Base url. The ODA instance is in public cloud and it cannot access the peoplesoft service hosted within VPN. Incorrect Port Number.	Setup a tunnel to expose the services in open internet. Provide the public url as the Base url in skill parameter.
	http status code 200 and system error	1. Incorrect Node name or application service name in the Base url configured in Skill. 2. Invalid Rest Target connector url (Peopletools → Integration Broker → Configuration → Service Configuration.)	
T8	Ensure Valid BotID, component and menu tagged to the Content Reference which is rendered as a chat client Tile.		

5 Additional References

5.1 Developer Guides

[PeopleSoft Chatbot Red Paper](#)

[Oracle Digital Assistant](#)

[PeopleTools Application Service Framework](#)

5.2 PeopleSoft Chatbot Videos

[PeopleSoft Chatbot Integration Framework Video Feature Overview](#)

[PeopleSoft HCM Absence Chatbot Video Feature Overview](#)

5.3 PeopleSoft Chatbot FAQ

[Frequently Asked Questions](#)

[Quest Video – Everything About PeopleSoft Chatbots](#)